

ActInf GuestStream 087.1 ~ John Boik "CogNarr Ecosystem: Facilitating Group Cognition at Scale"

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all right hello and welcome this is active inference guest stream 87.1 on September 5th 2024 we're here with John Boy and Lee ER so Lee how about please say hello and give some contextualizing thoughts introduce John however you like and then we'll have presentation discussion with John so go for it Lee thank you sure uh thanks Daniel so um uh I I I came across uh John through the active INF Institute because he was the only other person sitting on the strange intersection that I'm sitting on in this space between looking at societal transformation and active inference as well so I was I was very excited and um I joined the live the first live stream that he did on his science-driven transformation papers um and subsequently I contacted him and uh we both explored a kind of a shared interest I guess in uh functional forms of cognition at scale and how those can be facilitated and and John was talking about the the potential of some kind of software solution to facilitate that and uh the kind of the discussion that ensued over the following months um has led to a John creating this this paper so I'm I've kind of been in discussion with him sort of over the months and I'm really really interested to uh get more into this presentation okay thank you go for it John thanks Lee appreciate that but I think it was more like many months of discussions that uh we were engaged in that led up to this paper so thanks for all your input during that time um so my name is John boy I'm a u a research fellow at the Active inference Institute uh very pleased to be uh appointed that to that position I have two uh new uh preprints available to the public uh two papers that we'll be discussing today uh both on the topic of um of group cognition at scale as Lee mentioned uh I'm I'm proposing a a essentially software ecosystem online ecosystem for group cognition at scale and uh and we'll today we'll talk about kind of the some of the general ideas there the overview of that project and then uh on the second guest stream but maybe is this slides VI can the everyone see the slides right now and can you go to the next slide then please so there's two the two papers have two guest streams that we're doing uh today and then the next one is on September 10th uh and on the second one today we'll cover

some general general concepts and on the second one we'll get a little bit more into some technical details about the underlying meaning representation uh when we'll I'll discuss in a moment what that term meaning representation means but uh this first paper is available on archive U you can download the paper from there and um and let's get started next slide please so I'm proposing something called the cogar ecosystem and cogar stands for cognitive narrative and in concept and I and I should say this is just a concept this is this project is still in its incubation phase um uh the the two papers are the substance of what has been done so far on this project and I'm hoping that I can interest others in the research community and in other communities Civ Civil Society communities other communities to uh get engag with this project and help it to move forward so in concept this is an uh open- Source uh online extensible ecosystem of basically software but resources tools applications librar services that facilitates group cognition and especially for large groups um and and especially with regard to Del deliberation strategizing problem solving uh decision making and that kind of things but you'll see that as we go through these slides you'll see that there's really a variety of use cases for this but these are the use cases that I really had in mind as I was um developing the concept for the for the project so essentially cogar helps users to what I what the way I phrase it is tell their story and by that I mean to convey their beliefs to the to what to to their group about how the world Works relative to some situation so um a group can be anything uh could be a you know could be a um uh could be a a city could be a corporation could be a school could be any kind of group that is trying to you know process the the understandings and beliefs of its members towards some goal perhaps problem solving or some other goal so in towards that aim cogar helps users to tell their story that is convey their beliefs to the group about how the world works and I both I use the word story and I use the word narrative in the in the series the series of two papers but I don't I don't mean to confuse the reader because I'm not talking about a narrative in the normal sense in the like a written narrative where you write down several paragraphs or several pages of text it's this is a little this is a little different but I still call the representation of story because it's it's a story in the sense that uh that it conveys users beliefs about like why something happened and what might happen next and um what what should we do and what does it mean and all that kind of stuff so you know this the this belief really a belief model that the user is is creating for this situation maybe explains you know what is the explains what the problem is what are we trying to solve or here's my solution or what just I think this is what just happened or what did happen or here's here's what I think will happen next um here's what I here's what I think will the people that will be

affected by this issue uh I believe that so and so and so and so was involved uh I I think that this this situation means such and such and we should take you know to address it we could take these series of actions or I think I think this should be done or or we could learn this from the situation so it's a story in in the sense that it it it spans the who what where how why questions regarding some situation um next slide please so uh there's actually several sort of services and apps available for for for um group collaboration for example um uh voting apps and things like that um this the Cog our ecosystem is I would say a much more sophisticated uh uh approach to that and um it allows a user to tell what I call what I uh refer to as a potentially Rich story that that means the story could be long complex nuanced uh it could be conditional like for example I might believe that in some situation uh if if we do a and b then perhaps c will happen but if D happened first then I then something else is going on so it could be a conditional story um could be the could be uh the equivalent of maybe a few paragraphs of text very short a story or a story could be the equivalent of a book chapter you know it all depends on what the user wants to do so so how could such Rich information be processed by a large group uh uh that's the really the question behind cogar how how could how could that happen so the proposal is for includes a back-end computational system that does numerous things it assists users and in creating and constructing stories it helps the group to digest what all of these stories are saying and how they are the same and how they're different and you know what interesting points you know other stories make and where there's agreement and you know what are the what are the many ways that this issue is seen by this group of people um so it does like analytics on stories for example to help answer some of those questions uh the system also assesses stories for Quality like you know is this is this story uh on topic for whatever the group is trying to do whatever the whatever the situation is the group is addressing uh is it complete does it does the story just make sense in and of itself um and um and the the the system can use the that those kinds of Assessments to give feedback to the user like uh for example the a story might have some character in it named Bob and some other character named Sue and it might not be clear from what the user is cre in whether how those two people are related so the system might if the system is confused about what the relationship is it might actually ask them like ask the user like what is I'm not I'm a little confused what's the relationship between these two people so in that sense the system is acts very much like a like an Engaged active listeners like if you were home telling your story to a friend the system is kind of like that friend that is listening and trying to make sense of what you're saying and maybe asking you know pertinent questions to help direct your uh your

telling of the story so that it understands what's what what the story is about and the system also performs inference for uh uh for queries like perhaps other users other people in the group have a question for the for the Storyteller or the story or perhaps the system has questions like what's the relationship between Bob and Sue as I just mentioned so the system has to perform various kinds of inference to to help generate and answer those types of queries without without such a computational system the the you know the the imag Imagine A large group of thousands of people and thousands of people submit a long complicated story how would a group make sense of that if there weren't some kind of comput computational system doing assessments and inference and and uh guidance and assistance kind of behind the scenes next slide so I've been talking about a group uh a group that's interested in some some topic some situation or topic or problem but maybe we should uh take a moment and and just let me explain what what I mean by group so so we're talking about a group of people in this case and um and I I call I call a group a collection of people that demonstrate some kind of coord coordinated behavior in response to Dynamic conditions as as time elapses and conditions evolve so examples include just about anything you can think of that would be a a group so Social Clubs volunteer groups Civic organizations nonprofits corporations and even political entities like uh cities and states and Nations um a group can range from very informal to formal flexible to structured temporary to permanent small to large autocratic to democratic and the way it uh it uh manages information and manages decision-making and so a group senses it a group does many things you know generally a group has a purpose it form for some purpose and to carry out its purpose it generally senses its World processes information anticipates outcomes uh like the a group might say you know we we we should take steps A and B in order to be financially viable or you know some such thing it a group evaluates its options it makes decisions and then it takes actions uh either just within the group or within the world or in some way some kind of actions so in other words I'm saying that a group can be thought of as an organism an organism that performs cognition and just like any organism an animal a rabbit a human uh the animal has to do all of these things it has to sense it sense its World sense danger in particular look for food it has to process the information that it receives from the world and some of that information of course can be um uh you know noisy like it thinks it sees a wolf but maybe that's not a wolf what does it do what what should the animal do in this situation you know it has to anticipate the outcomes of its actions evaluate options make decisions and then take some kind of action in order to in order to maintain itself in the world and really in order to thrive in the world if that's possible so so I'm I'm thinking of

a group as an thinking of a group is an organism and that organism is just like any organism is performing cognition next slide so that that's a quick overview of cogar but uh now you know maybe it's a good point just to ask the question you know who cares or more in particular why should you care the The Listener um so uh obviously cogar is about group cognition um and maybe you see the need for improved cognition maybe maybe in particular you wish that some group perhaps a political group a nation or a state or or a corporation or some other group maybe you have a story to tell that you think would be important to help this group out to help the people to protect to to to warn of danger to highlight a problem or or a solution that you think no one else is thought thought of or is being underrepresented uh so maybe you have beliefs a maybe you have a story to sh share with a group and you want and not only that it's not just sharing a story because because if I I could write a a 50-page letter and send it to my congressman or you know some other some other group me group or group member and there's really no guarantee that that group will really understand what I'm saying in this in this thing you know that's um so you so whatever you share with a group you want them to you want the group to actually hear it to register it to hear it and then understand what it is that you're saying because why bother if uh why bother com submitting information if if it if it's not really heard or understood or maybe you maybe you look at the world as as I do and you see that the world has some serious uh you know the human world has some serious issues to solve climate change would be one example but there's many but pollution climate change poverty you know all kinds of social problems and uh and maybe uh maybe you feel that you know isn't there some way that we could do better isn't there some way that as a groups as as small groups as large groups as formal groups as informal groups as is there some way that we could cognate better that we could be uh more functional organisms group organisms and better a better make better decisions on our world and take better actions you know may maybe maybe you think we could do better or maybe you just are a scientist and you're interested just simply in the science behind cogar and and I would say that this project is large uh it's and complex and it touches or or could touch really just about every field of science uh I mean if you give me a scientific field I could could take a a guess as to how this cogar project might affect it or or how it incorporates ideas from that field let's just take one example say medicine like you might think like what what what do what what could what would cogar have to do with medicine because it's not a medical program you know in and of itself but doctors have to make decisions groups uh groups of doctors insurance companies other groups have to make decisions so that they have to come together as a group to make a decisions and

and assess some situation so that's one potential way another another potential impact on medicine is as a patient I have a story to tell I have a story about what happened um uh you know there was an accident I I I cut my arm the I stepped on a nail it was Rusty this and this you know this and this happened and I'm I'm afraid of this and this and I have a condition past condition of this and this and I'm allergic to this and this and all of that can just be written on a piece of paper you know as a like a regular intake form but but um in cogar that the story is standardized in a sense so that it's easily readable and understandable by humans and computers and your any a story that a patient is telling using a the cogar system would be um more accessible to inference and computation and um and there would be and also there's a natural record of of of stories in cogar so there would be potentially many areas of medicine as just one example of uh of how either kagar incorporates the science or could affect the science um or maybe you're just curious about uh you know what's possible for human groups you know we we live in a society that uh is faces all kinds of challenges uh there's future challenges on the horizon uh you know what what is the potential of society how how considering that a society is a group of people so again we're talking about groups how C how could How could a society best share information make decisions uh cognate really how could the organism called a society perform functional cognition or or or improved cognition over what is common or available today so I I I hope there's many reasons why you might care about cogner um next slide please so I've mentioned the term large group already so you know what does large group mean what does it mean to me dozens question mark hundreds thousands millions of participants and uh I would say the answer is yes all of those right um maybe dozens of people perhaps can fit in a room excuse me fit in a room and and talk freely about some situation some topic and uh and share Rich information verbally or you know in other ways face to face that might work with dozens but uh it doesn't work so well with hundreds uh and it certainly doesn't work so well with thousands or Millions so um as I said a moment ago cogar has many potential use cases but but one that I'm really targeting is the large group setting where really you know hundreds to thousands or millions of people are um are involved in the group and and having said that there's also use cases for just a single person using cogar there would be reasons why somebody might want to record their story in cogar just for personal reasons to as a history for example to look back on later and reflect when and there's other reasons to that we'll talk about shortly so so the question at hand is you know what kind of online ecosystem would allow say a 100,000 people to share some kind of Rich uh complex story and and in particular Dynamic story like I I I'm a member of a group

we're sharing our stories I I write down what my story is I I look at what you're writing I look at your story and I go oh that's a I like that idea I like that part of it or that's that seems really useful to me or oh I forgot I forgot yeah that I should have mentioned that myself so by sharing stories by sharing belief models we can come to um you know uh uh reach a you know better agreement our our belief by sharing our belief models can start to um influence one another and and de and the group then develops a shared belief model to an extent um and then and then it's not just about sh cogar is not just about sharing uh uh stories it's about using them in some kind of to fa to facilitate functional group cognition right so you can imagine a group Corporation any kind of group Civic group um a city state and the cognition for that group you might say is functional sometimes or really quite dysfunctional sometimes So Co cogar is is aiming at facilitating functional group cognition that's its purpose uh next next slide okay so um we've talked about what a group is uh and I've not and I've used the word cognition now many times so maybe we should talk about what is cognition um so in cognitive science one might come up with very definitions of you know what is cognition how does it work what is its purpose so I'm relying quite heavily on the active inference uh uh framework for the you know my definition of cognition and and the way the cogn system would work but active inference is uh is essentially a basian description of cognition it's a normative principle uh under writing cognition and and biological and artificial agents so in active inference there's an intelligent agent and a world that the intelligent agent is acting in maybe maybe there's even multiple intelligent agents acting in the world and an agent has an internal generative model about how the world Works uh that is it it it given some conditions um it sees that the that there's potential actions that could be taken and it and it sees that there's likely outcomes of various actions so uh an agent has a model of of how the world Works what would ha what will happen if I do a what will happen if I do B what'll happen if I don't run from that lion um and uh the the the the the the agent and its world are separated in a sense statist statistically separated by what's called a mark blanket and that blanket uh consists of agent actions and agent observations so as an agent I take some action that action affects the world the world reacts in some way according to the world's generative process and then things change conditions change and as an agent I perceive the my world but you know maybe it's dark and I don't perceive it very closely or maybe it's loud and I don't hear it very accurately or you know I get Sensations from the world but they're not necessarily I'm not necessarily observing the actual state of the world I'm observing some kind of noisy representation of that so uh so the the act the Markov blanket is a it's it's it's conditionally

separates the world from the agent uh and active inference is the framework that I'm using to to talk about cognition uh and and not just cognition how it works but also what is the purpose of cognition what is what is functional cognition versus dysfunctional cognition NE uh next slide please so what is the purpose of cognition well from an active inference perspective it's we can think of it you know simply as an idea realiz four part four-part CYCC process of I I predict what will happen if I act I I choose an action from my generative model I I the world changes I sense whatever however the world has changed and I learn from processing those sense sense inputs for example uh suppose I'm a young boy and I'm learning how to catch a baseball right if I don't lift my arm high enough I maybe miss the ball so that happens a few times I I learn that I really need to lift my arm higher if I want to catch that ball so uh so you know again uh informally and simply it's the cycle of predict act sense and learn and an agent does this it it engages in cognition to reduce its uncertainty be un certain about whether it's going to maintain achieve and maintain those conditions that it's thinks will be good for it that will that will help it's thriving as a as a young boy learning how to catch a ball I I want to catch the ball that will be good for me if I can learn how to catch a baseball and I want to I want to reduce my uncertainty that I'm going to catch that ball I'm going to practice hard and I'm going to learn from my mistakes and I'm going to reduce the uncertainty that when I put my glove up that ball is going to land on it or or I'm GNA out outsmart the Lion and get and get away from danger when the lion appears uh so an an agent favors those choices those actions that are and G that are going to ensure the greatest resolution of uncertainty under the constraint that preferred outcomes are realized that's the way friston phrases it so so you know so you know it's it's not just that I want to learn how to catch a ball or run from a line it's that I want to sure or pretty sure that I can catch that ball or or get out of the way of that line and uh and from all this I would say that active inference offers a potential description of functional versus dysfunctional cognition if if if I have a pattern of making poor predictions and therefore acting in ways that are that are harmful to me I you we could call that dysfunctional cogn if if I fail to remember what happened last or fail to remember how many times or the many ways that I missed the ball uh that could be also called dysfunctional cognition and cognition itself rests on many core uh functionalities like remembering and attention and and uh and communication and other other you know processes and any one of those all of them together and anyone in in individually could be functional or dysfunctional in some you know cognitive event uh next slide please so so given all of that background what what is the cogar ecosystem really so I'm G to I'm going to I'm going to claim or offer that the

that cogar could be viewed as a cognitive architecture so every organism has some kind of every organism cognates whether it's a amoeba or a human and um cogn cognition requires some sort of cognitive architecture uh for humans we have a central nervous system and and we also have computers and external tools and things that can be thought of as part of that cognitive architecture and uh the cognitive architecture essentially integrates information and and helps us coordinate our actions in the world and it's the and it's the set of components by which and through which cognition occurs and for a group that includes uh the a group as members so it includes the cognitive architecture of the individual members but it also includes the the the the set of norms that a group uses its rules its mechanisms and institutions its tools it's its sensors perhaps we're talking about a group that is a nation and it has sensors on air pollution you know something like that uh it's analytic processes its systems procedures all of which uh facilitate or guide cognition really make cognition possible and every part of a cognitive architecture can be functional or dysfunctional right so for a group say um maybe it's rules maybe it has certain rules that that are either enhance functional cognition or or detract from functional cognition so I I view cogar as a component of the cognitive archit architecture of the groups that are using it and again its purpose is to facilitate functional cognition versus dysfunctional cognition uh next slide oh lee yeah yeah hi John um sorry this is really interesting because um you know based on the discussions we've been having uh I'm kind of using the concept of cognitive architecture in some of the research that I'm doing in organizations at the at the minute and um and Tred to feed it back in So if you take a um an organization like a public sector body or a a commercial organization um would you would you consider that the the the structure of the processes are all part part of the the cogn cognitive architecture of that particular entity well I mean I suppose depending on how broadly you want to define a an architecture I suppose it can include just about every aspect of a group you know potentially but I think in a practical sense you could probably focus on those components of group type of you know those mechanisms and structures of the group and its dynamics that would be particularly aimed at uh sensing the really the four steps that I uh discussed just a minute ago U taking action learning uh you know sensing the world uh you know one of the things I guess one of the reasons I ask is is one of the things that uh tends to the the context in organization has a an enormous influence in shaping people's behavior um but uh because there's a there's a general kind of widespread view that cognition is localized within people rather than a kind of a you know an organismic view of it I guess where it's kind of distributed between people um in a hierarchical organization a lot of those factors if I mean

I'm almost viewing it now as a as a generative model but there's no way that certain processes that determine Behavior can actually be subject to change because they're not considered as part of the cognitive structure if you if you see what I mean so I think the idea um you know that that um groups could view themselves as organisms is really interesting but currently there's kind of more of a mechanistic kind of view of of organizations and I think that potentially gets in the way anyway maybe this is a discussion for a for a different no no no no no no no it's actually good because I I I I I wanted to emphasize the importance of this view you know thinking of a group as an organism as a cognitive organism and thinking of decision making and and those operations as aspects of group cognition so you know uh the world is kind of crazy in some ways politics are engaging for many people and this is just one example in the in the for political entity but you know we we we rarely ask ask what a u what a economic system is supposed to do or what a governance system is supposed to do or or really what an organization is supposed to do right so so what we're framing all of that now in terms of cognition and we're framing it in such a way that it is amenable to Scientific you know examination of whether this is functional or dysfunctional and and we're and and from an active infer inference perspective cognition has a purpose so so then we can start talking about is this organization the you know it structures the way it the way it's put together its rules its mechanisms its institutions fun is that functional yeah and and could right and could it be improved like now that we know what we're after now that we have the problem you know the the issue in Focus yeah how could we improve the the cognition of of uh I just give you a tiny little tiny little vignette so in this particular organization they have a performance management system uh but the so that if you if you to take the view of functional cognition you go the purpose of the you know what's the purpose of the performance management system what it's there to motivate employees reward them you know make them feel excited about their work but it doesn't do any of that what it actually does is it demoralizes them annoys them gets in their way and just generally pisses them off I mean it's actually doing the opposite of what's intended you take the whole system out and the organization would be better off but there's no it's not subject to any prediction error signal because it can't be no ever defined it in terms of you know purpose and outcome so it's a completely dysfunctional system that's just stuck there that you can't take out and nobody you know nobody has a view on it it's it's really weird but I think would really get to it but anyway sorry to interrupt the no no no no no no it's it's good it's good I I just want to emphasize again that this is a in a sense it's a change in world view it's a this is a comprehensive and potentially very useful way to con to understand what a group is

whether that group is a small informal group or a large formal group and and then to assess you know how well it's doing and how could how could we how could it do better right so this that's the question for cogar that I raised just a minute ago is like okay given the setup given our goals given the purpose of cogar so what kind how should cogar be designed you know like how should it how should it be designed to do what we're saying it should do you know that's the question here um but anyway but getting back to the back to where we were at in the slides uh cogar is I I would say cogar is a part of the cognitive architecture of the group that's using it next slide so actually this this slide speaks lead to some of the some of the things that you were raising there and and it speaks to some of the um some of the things that I just mentioned but so consider a society as a group let say a nation as a group again a group that is that is an organism performing cognition for some purpose and that purpose obviously for any organism is to thrive to to maintain viability to to do well to Live pass on information um so in the I I've wrote a series of uh of three papers a couple of years ago in 2020 and 21 2021 and I think maybe there's I don't know Daniel four guest streams for that series I think three or four or something like that that are in the archives so I encourage viewers to check those out too um and that series of three papers again looked at cognitive architecture only in this case it looked at uh societal systems as the cognitive architecture of a society so by core societal systems I mean for example e economic systems Financial systems governance systems legal systems Educational Systems you know these constitute a considerable aspect of the cognitive architecture for society we use these systems to sense the world to think to make decisions to allocate resources to take a variety of actions uh that that hopefully actions that better the you know the nation at hand so um societies around the world or nations around the world have I should really say Nations around the world have some kind of the core societal systems say economic systems that have some kind of particular design to them for financial system as a particular design but the question I raise in this paper is in the series of papers is out out of all conceivable designs you know existing designs or completely new designs which which of them might be most fit for this purpose of group cognition which of these might best serve the common good and and and obviously making good decisions that you know favor the vitality and well-being of a nation is serving the common good so out of any conceivable design for economic systems you know what is money for example is there some other way to think about money or some other alternative to money or how do votes happen or you know like any aspect of a core societal system one could imagine many conceivable variations on on a theme or just completely new designs new new ways of looking at what an economic system is and

from this umbrella of considering all of these systems as part of a cognitive architecture then one can then one can ask you know really what designs are fit for purpose and how do we of course how do we measure Fitness and you know what kind of information goes into assessing the fitness of some system but it opens the door for a science-based design of the systems by which societies live under and that is not the way that is not the you know our current system economic system Financial systems governance systems they were not you know developed through some kind of science-based process they were they kind of evolve through social social pressures and social conditions and and you know ideas from individuals and but we could revisit all of that um and and ask the question you know what what what designs actually would serve Us best and in the series I propose an R&D program to really to address that very question not just not just you know the not just as an academic exercise of you know let's write some interesting papers about you know how how governance systems could be redesigned the idea in this uh series is how do you what kind of designs would potentially be most useful and then how do you test them actually practically test them Implement them use them you know gain information from you know the the experience of of using them and learning and learn from them how how how could that happen how could that actually be done how could some kind of transformation you know I call it transformation in the series but how could some kind of transformational step occur and be viable and practical so in the in the series I offer you know I what I think is One path towards that but this current uh uh series of two papers that we're discussing today this is an it's an extension of the same idea it's again looking at the cognitive architecture of a group and asking the question uh in this case for cogar what kind of design of cogar would could best help uh group cognition and how could cogar be used by groups to you know to help them perform better cognition next next slide you know now that I've said next slide but there's just one more thing I want to say about the last slide uh uh you as I was about to say a few minutes ago lots of people are interested in politics and you know what a government should do and and they have ideas for uh you know like maybe somebody thinks that uh that um term limits on a for judges or for politicians should be limited that you know that that would be a you know that might be a solution to some of our problems or you know there's all kinds of of um proposals floating around for how to improve economic systems or governance systems or you know any any of the core societal systems and and even say corporations do the same thing you know they wonder how could we improve our Communications and all that kind of stuff but this concept this this this this really powerful approach of viewing systems as a as a cognitive architecture

allows those questions to become you know meteor in a way you know more relevant more with a greater high level understanding of what that system is actually supposed to do so I I believe that by viewing the the societal systems of a society as a cognitive AR architecture and looking at groups as as you know small or large groups as uh as as organisms that perform cognition I think that opens up all kinds of doors and avenues for investigation into what kinds of designs what kinds of you know rules and and approaches and methods and tools might actually uh help groups perform better cognition and and better solve their problems okay next slide so um everything we've talked about so far has just been sort of high level what is cogar what is it do what is its purpose but uh this slide really gets more to the to the question of how does Scot work you like under the hood so I I I've already said that I use the word story but I don't mean story in the regular sort of narr written narrative sense uh uh a user creates a story excuse me creates a story but and and it creates a story May perhaps by speaking you know in one version of cogner maybe a user speaks the story and the system hears it and understands it but maybe the first version of cogar would be uh somebody the user inputs a story kind of Passage by passage you know sentence by sentence and um the system takes that input those little passages and other metadata associated with passages and constructs uh a fragment of a of a of a story and that story is represented representative represented has a graph so it has a its internal meaning represent representation is called a is actually called a story graph and that story graph the reason that it's not s stored in just regular English text or text of some other language is that this representation allows is is more amenable to computation and could be better understood in the large group setting that in that sense there's other reasons too we'll get to but but the important thing for this slide is that is that actually what the user is creating is something that resembles a graph a graph of how the elements of the story and are are are uh are are related and story graphs are really the core Innovation on which Kar is based and that it it's that representation allows cogar to curate understand reason with perform inference for uh stories and these stories then represent really high quality information uh much of the information much of the vagueness of natural language is eliminated in this way not all the vague vagueness but some of the vagueness is limited in this way so uh at the top of this little graph you can see that a user types in for example text text passages those are translated by the system into what I call a story graph fragment just a small little graph for that passage and that story graph fragment is then merged with the growing larger story graph and um once it's complete that complete that the complete story graph can then be translated and used in various ways it could be translated back into natural language so that potentially a user could

listen to their story back like you know I I typed in this I typed in my story about you know what should happen for this particular situation and uh I could ask the system to replay that back to me and hear it perhaps uh it ALS this the story graph can also be translated or used as input into a a variety of inference models so probabilistic logical or other kinds of models and maybe it would be maybe it could be translated into alternative meaning representations that would be useful in certain kinds of inference models so that's what happens under the hood you're constructing in a story graph and you can view the user can view aspects of that graph as it's constructed to verify that the system actually understood correctly what I'm trying to say uh next slide please oh uh well fine this is good um so this is an example of a meaning representation a very common meaning representation called the abstract meaning representation for a particular sentence uh sentences a similar technique is almost always is almost impossible to apply to other crops such as cotton sowing beans and rice so so you can see that this sentence the the representation is structured uh there's rules for how the representation is constructed and there's a structure to it and certain links arrows lines between uh nodes have you know a given meaning um and um I I I have this in the paper and here just really just as as an example of what a what I would call a relatively simple uh meaning representation might look like so the cogar meaning representation is going to be like this only far more sophisticated um in in many ways so for just as one example um anticipation and predictions are an important very important part of cognition as we've already discussed and the meaning representation has to be you know sophisticated enough to handle the predictions of a of a story of a belief system like if if the uh if if the topic at hand is uh immigration um I have a beliefs about you know what kind of rules the immigration Authority should have and how it should be done and you know build a wall or don't build a wall this just one of the topics of of conversation in the in the papers these days um um but for my story uh there would be naturally would be uh anticipations and and predictions related like uh uh I if we build a wall then it will have such and such an effect if we don't build a wall it'll have such and such an effect if we allow Freer immigration it'll have such and such an effect these are all anticipations predictions and the meaning representation has to be sophisticated enough to handle all those and they're not just and they're not just um predictions they're probabilistic predictions no one knows what's going to happen in the future so all of those kinds of statements are are probabilistic so even even you know it has to be even more the meaning representation has to be sophisticated enough to handle probabilistic uh predictions about you know what a story means and what might happen next so uh this is a very simple graphical

version just to give you the user a very very high level rough idea of what's going on under the hood but in the actuality the cogar meaning representation would be much more sophisticated next slide so um what is the purpose of the story graph some of this I've mentioned uh already but uh it represents a story in a format that is less ambiguous than natural language natural language can be very ambiguous very U it the story graph is meant to be readable and understandable both by humans and computers so a human can look at a story graph or pieces of a story graph views of a story graph and uh and and and verify that the system understood the story correctly for example but computers then can also read and process the stories and then you know perform analytics and inference and do do the many things that the back end the computational system has to do the stories obviously then provide a structure for inference and Analysis and they also serve as interesting lingua uh to you know a natural language independent representation that's suitable for translations into other languages or uh uses inputs to analytical models or other things so that's what a story graph should do and then one of the questions is well what what what does the cogar story graph you know what is its design you know what is the what are the specifics of its design and that is a topic that I raise in the in the second paper that we'll talk about a little bit in the next uh guest room next slide please so um we've talked about stories and I've I've said a little bit about what what they are but let's maybe recap that just a bit so you know a user has some story in their head about a situation some you know issue or problem in the world and um and the user has an internal belief model about how the world works relative to that situation or problem and a story is a is a is an externalization of of that belief model so and that and it and it potentially is a dynamic externalization you know it can change as the user sees other stories or reads what other people think um so it it externalizes a user's belief system and makes it sharable you know that's the purpose of cogar is to share these belief models with other members of the group uh so again it the story conveys a user's beliefs and the the emphasis here is on beliefs right because um the the computational system is not going to predict what's going to happen in real life if the group takes actions a versus action B for example uh and it's not going to the system is not going to verify that somebody's story is AC is accurate is realistic or or is practical for the situation at hand those are those are assessments that the group of users has to do you know together and the system helps that process but it doesn't perform those kinds of uh assessments so so the system the system doesn't have to know the rules of gravity or you know Newtonian mechanics or it doesn't have to predict what's going to happen in the world it it re all the system needs to do is to capture a user's belief model and to communicate

that model with others and to help others help the group understand that model and how it compares to other models and um what elements are in it and you know what is its time span and you know what what is what what is this you know what are the characteristics of this user's belief system and and what is the user predicting will happen so it's it's the story just captures the user's beliefs right and then the Group shares and acts in some way on those beliefs so so that's it the the system helps a group to understand what its members believe that's that's really the core of this and then those by sharing those beliefs users can uh influence one another perhaps you know like in a setting next slide please all right so uh so we've talked about what a story is how it's represented of an under the hood what it's what the purpose of cogar is what the purpose of a story is and now we can talk about how this cogar system might be actually used and I've said that the emphasis is on you know the cognition in the large group setting and particularly for things like decision making and and collaboration and things like that but in actuality there's a very wide span of use cases that are possible with this kind of ecosystem um and and I'll cover just a few possibilities but I but these are non this is a non-exhaustive list there's many many possibilities and the communications in in its different uses you know may maybe there's an app developed for each particular kind of use but so maybe there's many apps that are part of the ecosystem but the but this but the sort of events I I call it a group event a group a group defines an event and then invites people to participate in the event so for example a city might if a city is using the system it might um um Define a group event to solve some particular issue maybe a trash issue or recycling issue or a road issue or some other issue and the and the group would Define the topic for the event and the purpose of the event and the rules for the event and the methods for the event and how decisions we made and you know the setup for how this event is going to unfold and then it asks for a group input through story graphs so depending on the event and the group and the purpose of the event the communications can be many to many as in you know collaborative decision-making for example they could be many to one or even one to many so uh and and and and I've been emphasizing group use but again an individual might use this just for their own personal use just to record their their understandings of the world at a certain time for example so some possible use cases are in decision-making such as by a corporation or decision uh Corporation or city uh maybe customer service uh emergency reports public polling instructions for how to put together bookshelves or instructions for how to run a machine for example uh system descriptions of you know what is what is this system and how does it work or maybe there's whole variety of educational uh use cases also and uh later

versions of cogar could be could include things like a role playing or game playing like uh intentional games uh purpose games and strategizing what what what here's here's my belief system of the world relative to the situation and uh you know the people involved in the story or the characters involved in the story or you know so and so and so and so and so and so and what would happen if I if I take action a you know what's likely to be the outcome what is what is player b g to likely to do if I take action a so later more sophisticated versions of cogn Might include that kind of stuff where you user might play against the system or two systems in a sense might play against each other to as to to you know to to strategize for some situation um next please so the next couple of slides are just a few examples that I have in the paper and um for use cases so I'll cover them briefly uh um so I have written at the top of these slides some text and this isn't actually the text that you know somebody would uh write in a in the real codar this is just a snippet an example snippet of the kind of information that a user might want to convey right so this user there's an issue some issue X and this this this person believes that in order to address that we should first address issue why that caused X and why started when you know it's it when such and such happened and it had the effect of such and such and this person believes that if we don't solve issue y then some event a some unwanted event a is likely to occur maybe within a few months and then ex the original issue is going to be impossible to solve so if that if a happens then you know there's going to be all these effects maybe you know and there's some this this in the story there's some kind of uh you know uh 20% value put on the risk of this or or the or the outcome of this and uh this user says a possible solution to this this problem why is to do you know such and such and it's conditional so do such and such if event B occurs and such and such if event C occurs so there's some kind of you know complicated story conditional story involved here that has predictions in it um it's it's uh you know there some shoulds in it you know we should do this uh and how does how does how does all that work uh for the decision-making case well one common I I believe what would be a common setup for the decision-making case is um uh a group forms comes together for an event and individuals you know create their story that's kind of at the top of this diagram they create their story and then um you know submit their story and the system the middle the middle block though the system does its assessments and computations and you know the various things that the system is going to do and then it reports back to the group about the range of stories that have been submitted and what they include and who they involve and you know what the kinds of of solutions are have been proposed and the various kinds of summaries and details about about the the set of sto that have been involved and then

the group uh uh you know digests that information and looks at individuals can look at individual other the the stories of other individuals uh so users can kind of take in what the group is saying and maybe a user thinks oh that's that's you know Bob had something really interesting to say there and I you know I should really include that as part of my story that was a good idea I didn't include it so should included so um the next step is a this this process uh Cycles in a series of rounds and at the end of each round there's a kind of dialogue and digestion phase where everyone looks at what's been done and gets a better idea of what the group is thinking and maybe they adjust their their story uh you know accordingly and hopefully over time over a series of rounds the a proposed solution that the group favors starts to become clear right and then then this process can end when it's sufficiently clear and there's enough agreement that we we understand each other we we think we understand the situation and we think as a group that this you know this this this this solution or this set of solutions might uh be the best to actions to take and so that's the that's group decision- making might occur over a series of rounds uh next slide here's a an example use case for emergency reports this is a this is a person who wants to report that they just witnessed an accident uh there's an accident with some kind of truck of some kind uh maybe it was going too fast maybe it was because of the wet conditions it shouldn't have been going so fast uh lots of accid like side information there's been many accidents at that corner so maybe that that maybe the the uh maybe the local conditions have something to do with this but anyway the the truck had a telephone pole and now there's Sparks and the grass and uh there's a school and children are playing outdoors and um you know all of this is a little unclear to me I'm about a block away but um you know that's what I see and I report it so you know there's a there's an interesting story and and you could imagine even a more complex story of someone that wants to report report something maybe they want to report a crime for example or in this case an emergency report and there's also there's various kinds of inference that has to happen like under the hood for this right like if you read the the sentences that I that I wrote there I never say that a uh wire uh fell from the lines and is now on the grass that's sort of inferred right so there's various things that the that the the system has to infer about this story in order to make sense of it and I say that there's uh you know this user is saying that there's children present but the implication is is that there's a danger to those children right so there there's actually a forecast like a prediction like if we don't do something someone might get hurt so these children might be harmed you know because of the a the uh you know the aftermath of this accident um so uh so uh you know this is this is this is a this little story is a lot different than for example calling a

hotline and the hotline instructs you to press one if there's a fire press two if there's a you know something else and then if you press one for a fire it says you know press one if um the fire is in your house press two if you know whatever so there are other computations systems out in the world that seek to address seek to gather this kind of information you know for the purpose of making an emergency report but I'm claiming that out of all out of all the kinds of systems that I know of the cogar system is much more functional and has the capacity to convey a much richer uh story understanding of someone's understanding about about a situation and perform inference on it and record as a history the story and how it evolved and many other things that the that the computational system does offers next slide please um this is a uh this is an example for customer service this is an example of a of a U of A many to one sort of communication right in the sense of um many customers are are communicating with one agent one one Central agent which is the corporation in this case so uh uh people are are writing in and they're sending in their broken parts and they're you know they're they're giving a description of why this part worked or you know is this part defective or why was this why was this aspect of the part designs seemingly so poorly or could you fix this design or you know like here's what here's what's here's where it's broken this this part here this this Part F has a leak in it and it causes this other part H to slip you know some kind of sliding mechanism to slip and uh and you know seems to me that part was it wasn't really broken it's just that it didn't really fit so was that by Design or is this part defective or you know is what's what's what's going on here with this so you know this is Rich information that a that a corporation might receive about uh its products from users and uh you know much more useful than say just a you know a couple of sentences written in natural language or maybe a paragraph written in natural language because you know the corporation could use this to really assess you know really analyze uh closely uh uh what what what's going on and how many times does this happen and how many times have people complained that that part was too thin and you know all that kind of stuff so uh so there's there's an example of a of a of a many to one sort of communication scheme next slide this is uh also an example of them many to one so uh you know lots of political things happening in the world right now elections and so on and lots of polling going on too for the for the different elections and um you know typically in polls you might be asked a yes or no question you know do you agree with the statement yes or no or maybe you have a a a short um uh uh multiple choice you know do yes I agree strongly or no I disagree strongly or I can rate my agreement on a scale of one to five or you know some something like that but that is not near as rich as as in taking what the user

actually has to say and what what they what they think are the important aspects to to this to the question that's being pulled about in this case the user is is is there's a poll for some candidate and uh this user is saying you know I thought her past decisions on certain issues were terrible and I've been fuming about them and um I don't trust her on this issue because of such and such statement and I don't even think she's sincere on that issue and anywayss that issue could be solved by some other much simpler process which you know I'm I I mention here and uh I don't think I'm GNA vote for her so you know this is Rich information and again there's inference that has to go on behind the scenes with the with the uh system for example the word fuming you know that could has various meanings uh does it mean that the person the writer is on fire no not in this it means that they're angry so out of the various definitions of the word fuming the system has to figure out what definition is the user aiming for here um next slide and then I think we're done with the next slide uh where we come to the end of our talk so uh what is next for this project uh I've already mentioned that cogar is in the incubation phase and um and I put out the vision in these two papers a vision for it and um really it's a big project it touches on many aspects of science computational science complex system science all kinds of different Sciences it touches on and this is all beyond my capacity as one person to create so if this project is going to move forward it it will need your help uh whether you're a scientist or you're uh belong to a Civic organization or you're just an interested citizen or whatever there's there's there's some way that uh just about anyone could be involved in this especially as it picks up steam and and moves starts to move forward um there's many many interesting questions to address I I I raise a few future research questions in the papers but there's many many more That Could Be Imagined and I encourage people to reach out to me if they have interest in this project um upcoming there there's the guest stream on September 10th that's next Tuesday and um uh I I I believe I'll be starting a cogar project at the Active Inference Institute as people can get that involved that way and get involved with the Institute that way and uh they there should be some additional talks coming up there's a active Inference Symposium coming up that I think I'll present that and um and there's a few other talks I'll be giving to to private groups and and such so uh again please reach out to me if you have interest in this and uh and uh show up next time for the for the guest stream on September 10th and we'll continue the discussion awesome thank you John I will read some of the questions from lot chat upcycle Club wrote is group cohesion one of the goals of cogar um so good question and and you know the the in any kind of group there may be very different opinions of what should happen and um there may be lots of friction in that group right so um

it's not that cogar is necessarily going to solve all kinds of friction problems and get everybody on the same page for every setting you know that would that's not the way the world works right but but what it what cogar does do is a sets up a process and a means and a mechanism by which people can share their stories and perhaps come to some agreement uh shared understanding you know share the share their way they see the world and maybe the group will come to some kind of shared understanding if the purpose is decision-making you know that's kind of the context here so um hopefully a group will come to some kind of uh agreement but but but even if they don't cogar can cogar has a record of the stories and of the differences in stories and maybe there's really kind of two subsets in this group or three or four subsets in this group and each subset views the situation in a very different way well at the very least cogar is going to help the group to understand that there's you know different ways to look at this problem and different proposed Solutions and it's going to record the the the the evolving stories that are related to to this you know like how they how a particular story started and what it what it r to become over the course of cognition and um so you know maybe that maybe the group will actually come to a a well uh you know widely accepted uh decision on some topic or maybe it won't but at least it's going to have a functional process to you know to to go through to to understand itself right so that's the that's the that's the hopefully yes and but you know that what happens the Dynamics depends on what the group is and what the topic is thank you all right next question from abis tolge who wrote two questions one how does cogar maintain consistency in story graph updates when passages are the single source of record second question how are changes propagated across the story graph and Associated models without conflicts um can use can use um back in the slides to the to the uh process process of creating a story it's kind of far back there one more yeah there we go this one yeah there you go thank you okay so so one could imag you know we're thinking about how should cogar be designed you know how should how should it work you know what should it do and um you know one could imagine the a users could input text type in something and then that text becomes a story graph fragment here and then the story graph fragment becomes a story graph and then the re and then maybe the user realizes oh I I didn't mean that or I didn't make that clear I want to add something so how should the user edit this thing right you could edit the story graph or you could edit the graph fragment or you could edit the original text and then rebuild the story from there right so how should it happen well um the simplest approach and the approach I suggest in the in the paper is that all the editing really happen on the text passages because if if you uh allow edits on the story graph then they

have to propagate in this diagram have to propagate upward to so you know to change the original text passages so that everything is in agreement all the different sources of information are in agreement so it would be easier and simplest if there's only one source of information and that the entire graph can be um uh you know regenerated from that source as the as the information changes um what was the second question I think maybe I answered how how is the internal consistency and and the statistical consistency and then also the the changes of through actual propagation how does conflict how are changes propagated across a story graph and Associated models without conflicts yeah yeah that's that's that's I think I answered the question or at least I tried to um so there's only one source of record and that's the text passages itself but suppose now that the that the system uh suppose the user created a story and the system uh uh responds and says hey I don't I don't understand is is you know is Bob Bill's brother or his boss I don't understand you know what this relationship is or who when you say he which he are you referring to Bob or Bill or you know who who who is it that this this passage refers to so now the now you have a situation where um you know maybe maybe the is is um are they brothers or are they co-workers maybe that's the question the system poses and maybe the user responds their brothers you know as a response well how does that how does that response you know get into the story graph right because you don't want the system to edit the story graph because then the system would have to change you know propagate that changed information back up into the text passages and that might not work so well you know what if the system doesn't propagate it correctly uh the user could simply change their original text passages and and and make clear that what the relationship is you know what whatever the question is so you know that's a clear way us just changes the text passage itself but another way is there's some there's metadata associated with a text passage and that metadata might actually be you know include the kind of the conversation between the user and the and the and the system the system says the system asks you know are they brothers or co-workers and the user's response is they're co-workers okay well that that's meta information that is now you know saved as meta information and can be become part of the graph you know Downstream what about the human element of conflict like using Code as law legal code plain text repositories versioning and forking and instancing how does the social reality of the conflict not go into the fractal Linux kernel separation and what would actually you said one version of record and whether or not that resolves logical ambiguity that's an internal question what about the actual existence and production and power and influence over that one version of record If there really is one um are you asking uh how

reliable is a versioning system uh like if there's some kind of versioning system where the user you know changes information is there a reliable versioning system that can keep track of all these changes is is that yeah and even given a reliable complete perfect git type versioning of text how how does that not or does it or Etc have conflict outside of the text well I'm I'm not exactly sure what you're asking but let me pose this situation to you suppose that um I don't know um suppose that the that the system is unclear about what the story is saying and ask the user you know are is bills are these two people related are they co-workers and the user responds the user makes a response so therefore changes the record changes the metadata and says something really vague like uh they're they they they know each other that's the response that didn't really address the question right like it didn't resolve the issue that was raised right but then so what that means is that the story is simply vague right the the story is not a maybe not a great story it just has inconsistencies within it it's it's uh not logical maybe or it's just too vague to understand what it what is being said so that's not the is that's not the problem with it's not a problem of cogar itself that's a problem with the telling of the story and its completeness and its logical consistency and things like that and just like any just like just if you and I were in a room together and we're hashing out what we think about some situation I may I might make some vague statement to you about you know people involved and you might think in your head what what is what is he saying and you know like that's the way it is right that's the way it goes and you might ask me you know can you clarify that but maybe I give you some unclear answer and then you still don't know what I'm saying and you walk away thinking well John doesn't really know what he's saying but that's the that's also can happen here right so if I'm a user and there's some a group that I'm participating in and there's some event that I'm offering a story to and I and I submit some a story that is especially if I have a history of submitting stories that are unclear logically inconsistent incomplete you my reputation in that group is probably not going to rise very high you know people will eventually figure out that he he doesn't he's not clear on what he's saying you know or else he doesn't know what he's saying so you know those are that's that's part of the Dynamics interpersonal Dynamics and and the you know assessing assessing who in the group can be rely on to give a thoughtful uh you know reasonably useful contribution and who should we pay attention to like the group has an attention mechanism right naturally and maybe even facilitated by cogar to some degree and that attention mechanism would sort of naturally focuses on contributions from people who are known to provide clear uh useful information in the past or maybe people who have good predictive capacity like maybe I get a a Rec a record

and a reputation because I've participated in this group many times or on many different events and a lot of what I said was going to happen in the world actually happened you know I'm good at making forecasts of what's you maybe maybe I'm good at making forecasts so maybe the group is going to weigh my contribution a little bit more than some other person's contribution that was unclear or made poor forecasts so I maybe I maybe I kind of didn't answer your question question Daniel but yeah I I think it it touches on and you site and discuss some related projects related to like meta governance like you talk about modular policy and about attention economics and then also the trust networks it's uh lot of interesting work like John Ash's work in cognism and just thinking about how the attention distribution which is changing moment to moment based upon what kinds of message passes and and Quantum reference frames the interpretant deploys in that moment in the information environment setting and then how that relates to like formal and informal and all these heterogeneous systems which I think will bring us through the series towards active inference and what does active inference do Beyond you saying there's positive alternative narratives however such is some of the most common statements that people implicitly or explicitly say yeah well you have you raised many good points uh I'd like to talk speak to a couple of them uh one is that you know by by by thinking of you know a group decision-making as a cognitive process and thinking of a cognitive process as being a purposeful uh process and uh and and using using active inference and other uh uh Frameworks to potentially you know evaluate how maybe there's ways to evaluate how functional a given process was maybe in retrospect a group can look back at its history and assess how functional is our cognition how how how are we doing how how is information flowing how what are the many you know from a from a scientific standpoint what are the many ways that we could evaluate the process of cognition for its you know capacity or functionality or other things so you know those are some of the nitty-gritty scientific uh questions that that this project is sort of uh offering in a way uh another thing that you mentioned that I'm really glad you mentioned was meta governance and um uh it's an interesting group uh and I I you know cite them in the first paper at the end um but uh I did I did it reminded me that I wanted to say that that um you know cogar itself is sort of a group agnostic like different kinds of groups could use cogar a cogar app and that that the different groups might have different uh you know levels of formality or different ways they want to make decisions maybe if it's a decision-making uh you know process they're interested in maybe they maybe they want to use a majority vote maybe they want to use rank Choice voting maybe they have some other you know unusual system they want to use maybe they

want maybe the group is organized in a particular way maybe it's an autocratic group or you know or a very Democratic group or something right so cogar could be used by any group the group is going to Define what it is and how it works and what its rules are and you know how things go and then cogar seeks to be the structure on which then U you know uh Co cognitive event group events can occur right and that is really good because it doesn't cogar is not responsible for you know uh forcing all users to to use some particular method or approach or rule set or whatever you know there's room for Innovation within the cogar system there's room for experimental experiment experiment within the cogar uh system there's a history of of events uh that can be assessed in hindsight for how well they worked uh how did they how well did they guide a group towards uh shared uh you know widely widely shared and uh valued solutions to problems and things like that so uh yeah awesome okay another question from the live chat and anyone else can continue to write questions and we can continue to explore a few things synthetic data generator wrote can it be used as a protocol for group mental model harmonization well that's what it is you know if I understand the question correctly that's sort of what it is um if you and I are together in a room and we're talking about some situation in the world I have a model of in my mind about how how the world works and what'll happen if we do something or not do something and what it means and what it's what the future might be and you have the same you have a your own model in your head and we're and we're communicating aspects of that but what but what cogar does is it kind of makes that process available to the large group setting where there's potentially millions of people who are participating in this and what it what it what it does is it makes the model explicit right it it puts it down in an explicit way so that it can be assessed and analyzed and considered as whether it's different how different it is from other ways of approaching the situation or you know what its components are so um you know I I I that is what cogar does is it it represents a user's mental model of of a situation yeah and I think an important aspect of the harmonization ation is whether or not any given sense of harmonization is achieved or whether that's like a positive attribute used like coherence informally by taking like a dynamical systems parametric modeling basian probabilistic there actually are ways like using fmri analysis on the brain that rhythms of different time scales different Mutual information different Kale divergences like there's a whole toolkit of harmonics and Analysis so this is one mapping strategy and representation however it doesn't um need to be the only mapping or representation one that has some great causal properties though and helps be like kind of a repository format for the perception cognition and action that make things about a given

system specific rather than like the generic active inference formalism yeah there's a recently there's been uh several papers published on the how groups harmonize their belief systems as they come together to you know understand the world or make decisions so um there's a already prior work on uh what kind of what harmonization in a group setting might look like and how that works and how it might be ma even mathematically approached you know for modeling purposes yeah and another another thing that you just raised there a moment ago Daniel is that is the idea of a repository so suppose that cogar existed and were you know say millions of stories in cogar and you know hundreds of thousands of group events that could be analyzed in retrospect think of the value that would be to science for understanding cognition and understanding uh you know how information is conveyed and used and flows and you know U what kind of designs for rule systems or uh inputs or whatever facilitate uh nition in the best way I mean the just the existence of a repository of externalized belief models of you know complex externalized belief models of a variety of individuals on a variety of topics itself would be extremely valuable for cognitive science and computer science and various other you know psychology other various other aspects of science yeah this relates to a question that synthetic data generator followed up on it's an interesting question Lin and Simon were the seminal researchers in this diagrammatic reasoning is the explicit model visual or linguistic is the um um is if I can interpret that the question is the story graph visual or linguistic is that about it um uh so it's it's a comp the story graph one of the purposes of the story graph the meaning representation is to allow computation so um in that sense I guess you would say it's linguistic it's it's it's trying to um record the trying to capture the meaning of what a you know a phrase is what the user is trying to say right and how that phrase relates to other phrases of what what the user is trying to say but it's also visual in the sense that suppose and I'm glad you brought this slide up but suppose I'm I I'm creating a story and now I want to check whether the system is uh understanding that story correctly right so what I might want to do is I might want to view a portion of the story graph under the hood and see if the system put those things together correctly so that aspect could be sort of visual as a user I could look at something you know like this like what you have on the screen and go like okay I see that u i see that these two concepts were properly connected with each other it actually you know Sue actually is related to Robert in uh you know this particular way or it was it was Sue that dropped the gun not Bob or you know whatever the story is so a use there might be visual tools that the user would have available in order for them to you know query the query the underlining story graphs or even query someone else's story graph you know there might

be visual tools for that and and a result from a query might be a visual result so that you know it sort of makes sense to how the different elements are connected with each other but the but the but the you know kind of the core thing happening under the hood computation uh based on the linguistic meaning of of of phrases and and the intent of the user yeah a lot there especially programs considered and graphs considered as diagrams like from an applied category theoretic perspective the data structures themselves already being diagrammatic symbolic now they they might not be the right semantic representations for given system with this available observable with this unobservable model for this person with this accessibility that language like Etc but that is a cognitive situation with the human computer interface so whether it's a phone and uses this interface or uses another interface part of the interesting question with a generative model interoperability is can those different blankets different screens be also connected to a system that has an underlying representation that those perspectives come filtered through right right right right you you you raised um the field of Applied category Theory there a minute ago which is really interesting lots of applications of Applied category to this sort of setting because maybe you want to translate this graph into some other representation uh and you want to make and you want to uh you want that that um translation to preserve meaning well that you know that that's a funter in in um in applied category Theory so there might be ways to use applied category theory in in if even designing the structure of this you know the under the kind of like what structurally how does a story graph work how what is it how is it stored how is it retrieve from a database and you know how is it manipulated and how is it you know what kind of functors translate a particular story into a different version or visualization or or you know um really just a different you different version of that story or sub substory or translated from a story graph into English or a story graph into animation or you know like whatever the case whatever the case might be awesome well this was a great first go fort one maybe just resta what are we going to head towards and do to what would be fun and what what for those who are watching now could do between now then yeah well um uh so we've today we've kind of talked about the kind of the general highlevel what is cogar and what is it supposed to do uh next time when we get together we'll talk more about meaning representations and you what what kind of you know what should the cogar meeting representation look like what should it do what what characteris s should it have how was it put together how does maybe we'll have a chance to talk about how does applied category fit into all of this and um um in the meantime uh you know I W I encourage Watchers to take a look at the papers and uh and also if there's questions um you know feel

free to send questions I think I can be reached through the Institute somehow perhaps and
um or post question and maybe we can try to pick up any of those questions or comments or
suggestions you know that are offered in the intermedium intermediate part pick it up next
time okay awesome thank you again John and Lee too so see you next time sounds good bye e